Matter Waves and the Schrödinger Equation

PHYS 2203 — Introductory Physics III

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Overview

Last lecture showed that electrons, like photons, travel as waves and are detected as particles, with a de Broglie wavelength $\lambda = h/p$. We begin by using this wavelength to find where the dark fringes fall in an electron double-slit experiment. We then write the wave–particle relations in terms of the wave number k and angular frequency ω, and see why the wave speed $f\lambda$ is not the particle's speed. We then ask what equation a matter wave obeys. Comparing with waves on a string and electromagnetic waves leads to the free-particle Schrödinger equation, whose solutions are complex. We close by interpreting $|\Psi|^2$ as a probability density and examining the simplest solution, the plane wave.

1 Properties of Matter Waves

A matter wave is described by a **wave function** $\Psi(x,t)$, also called the probability amplitude. Like any wave it has a wavelength, a frequency, and a speed.

Wavelength and frequency. The de Broglie wavelength comes from the momentum, and the frequency from the energy, with the same relations as for photons:

$$\lambda = \frac{h}{p}, \qquad f = \frac{E}{h}. \tag{1}$$

Example: an electron double slit. Electrons moving at 3.0×10^6 m/s pass through two slits 0.020 μm apart. The momentum is the relativistic $p = \gamma m_e v$ (Lecture 17). Here $v/c = 0.010$, so

$$\gamma = \frac{1}{\sqrt{1-(0.010)^2}} = 1.00005 \approx 1, \tag{2}$$

and $p = m_e v$ to excellent accuracy. The wavelength is

$$\lambda = \frac{h}{\gamma m_e v} \approx \frac{h}{m_e v} = \frac{6.63 \times 10^{-34}\ \text{J}\cdot\text{s}}{(9.11 \times 10^{-31}\ \text{kg})(3.0 \times 10^{6}\ \text{m/s})} = 2.4 \times 10^{-10}\ \text{m}. \tag{3}$$

Recall from Lecture 8 that for two slits a distance d apart, the waves from the two slits interfere destructively (dark fringes) where the path difference $d\sin\theta$ is a half-integer number of wavelengths:

$$d\sin\theta = \left(m + \tfrac{1}{2}\right)\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \tag{4}$$

The first minimum ($m = 0$) is where the path difference is half a wavelength:

$$d \sin\theta = \frac{\lambda}{2} \qquad \Longrightarrow \qquad \sin\theta = \frac{2.4 \times 10^{-10}\ \text{m}}{2(2.0 \times 10^{-8}\ \text{m})} = 6.1 \times 10^{-3}, \qquad \theta = 0.35^\circ. \tag{5}$$

Even with slits only 20 nm apart, the fringes are very close together. This is why the electron double slit was not achieved until the 1960s.

Wave number and angular frequency. It is usually more convenient to work with

$$k \equiv \frac{2\pi}{\lambda} \quad \text{(wave number)}, \qquad \omega \equiv \frac{2\pi}{T} = 2\pi f \quad \text{(angular frequency)}, \tag{6}$$

and the reduced Planck constant

$$\hbar \equiv \frac{h}{2\pi} = 1.055 \times 10^{-34}\ \text{J} \cdot \text{s} = 6.58 \times 10^{-16}\ \text{eV} \cdot \text{s}. \tag{7}$$

The wave number is a "spatial frequency": the number of radians of phase per meter. In these variables the wave–particle relations become

Fundamental wave–particle relations.

$$p = \frac{h}{\lambda} = \hbar k, \qquad E = hf = \hbar\omega.$$

These hold for all particles, massive or massless.

Wave speed. The speed of the wave crests is

$$v_{\text{wave}} = f\lambda = \frac{E}{h}\frac{h}{p} = \frac{E}{p}. \tag{8}$$

For a photon, $E = pc$, so $v_{\text{wave}} = c$, as expected. For a massive particle $E \neq pc$, and E/p is *not* the particle's speed. The speed of a matter wave turns out to be of limited use. The speed that matters, the speed at which a localized particle moves, is a different quantity, the group velocity, that we will meet in Lecture 28.

Photon formulas do not apply to massive particles. For a photon, $E = pc$, so $E = hc/\lambda$ and $p = hf/c$. For a massive particle $E \neq pc$, so

$$E \neq \frac{hc}{\lambda}, \qquad p \neq \frac{hf}{c}.$$

To find the wavelength of a particle with a given kinetic energy K, first find its momentum (for example, $p = \sqrt{2mK}$ for a slow particle), then use $\lambda = h/p$.

2 Wave Equations

Every kind of wave obeys a **wave equation**, a differential equation in x and t that its wave function must satisfy. Two familiar examples:

Waves on a string. From $F_{\text{net}} = ma$ applied to a piece of string,

$$v^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}, \qquad y(x,t) = A\sin(kx - \omega t), \quad \frac{\omega}{k} = v. \tag{9}$$

Electromagnetic waves. From Maxwell's equations,

$$\mathbf{E}(x,t) = A\sin(kx - \omega t)\,\hat{\mathbf{y}}, \qquad \mathbf{B}(x,t) = \frac{A}{c}\sin(kx - \omega t)\,\hat{\mathbf{z}}, \qquad \frac{\omega}{k} = c. \tag{10}$$

Matter waves. The wave equation for matter waves is the **Schrödinger equation**. For a free particle (no forces, no potential energy) moving along x, it is

$$\boxed{-\frac{\hbar^2}{2m}\frac{\partial^2 \Psi(x,t)}{\partial x^2} = i\hbar \frac{\partial \Psi(x,t)}{\partial t}} \tag{11}$$

It cannot be derived from anything more basic. Like Newton's laws or Maxwell's equations, it is accepted because its predictions agree with experiment. Below we will see that it is at least consistent with the classical energy of a free particle.

Unlike the other two wave equations, it is only first order in time. This is necessary. With one time derivative, the total probability of finding the particle somewhere stays equal to 1 as Ψ changes in time. A second-order equation would not keep the total probability fixed. The price is that space and time are treated differently, with two derivatives in x but only one in t, just as $E = p^2/2m$ has p squared but E to the first power. In relativity space and time must enter on an equal footing, so the Schrödinger equation cannot be relativistic. Relativistic wave equations, and what they reveal about spin and antimatter, are the subject of Lectures 39 and 40.

3 Complex Wave Functions and Probability Density

The Schrödinger equation contains $i = \sqrt{-1}$, so its solutions are **complex**:

$$\Psi(x,t) = \operatorname{Re}\Psi + i\operatorname{Im}\Psi. \tag{12}$$

This does not mean matter waves are unreal. A complex function is just two real functions, its real and imaginary parts, and the Schrödinger equation needs both. The wave function itself is never measured directly.

Complex numbers. Recall the complex conjugate and the magnitude,

$$z = a + ib, \qquad z^* = a - ib, \qquad |z|^2 = z^* z = a^2 + b^2, \tag{13}$$

and Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta, \qquad |e^{i\theta}|^2 = \cos^2\theta + \sin^2\theta = 1. \tag{14}$$

Probability density. For light, the probability of detecting a photon is proportional to the intensity, the square of the wave's amplitude (Lecture 24). For a matter wave the amplitude is the complex number Ψ, and its squared magnitude includes both the real and imaginary parts:

$$[\mathrm{Re}\,\Psi]^2 + [\mathrm{Im}\,\Psi]^2 = \Psi^*\Psi = |\Psi|^2. \tag{15}$$

If the wave function is nearly constant over a small region, doubling the width of the region doubles the chance of finding the particle there. So $|\Psi|^2$ is a probability *per unit length*:

Probability density.

$$\text{probability density} = |\Psi(x,t)|^2.$$

The probability of finding the particle between x and $x + \mathrm{d}x$ is $|\Psi(x,t)|^2\,\mathrm{d}x$.

4 The Plane Wave

The simplest solution of the free-particle Schrödinger equation is the complex **plane wave**

$$\boxed{\Psi(x,t) = A\,e^{i(kx-\omega t)}} \tag{16}$$

where A is a constant. To check that it is a solution, take the derivatives:

$$\frac{\partial^2\Psi}{\partial x^2} = (ik)^2\Psi = -k^2\Psi, \qquad \frac{\partial\Psi}{\partial t} = -i\omega\Psi. \tag{17}$$

Substituting into the Schrödinger equation,

$$-\frac{\hbar^2}{2m}\left(-k^2\Psi\right) = i\hbar\left(-i\omega\Psi\right) \qquad \Longrightarrow \qquad \frac{\hbar^2k^2}{2m} = \hbar\omega. \tag{18}$$

The plane wave is a solution for all x and t, provided k and ω are related this way.

The connection to classical physics. With $p = \hbar k$ and $E = \hbar\omega$, this condition becomes

$$\boxed{\frac{p^2}{2m} = E} \tag{19}$$

Since $p^2/2m = \frac{1}{2}mv^2$, this says the total energy of a free particle equals its kinetic energy, as it must with no potential energy. The Schrödinger equation is built on the classical accounting of energy.

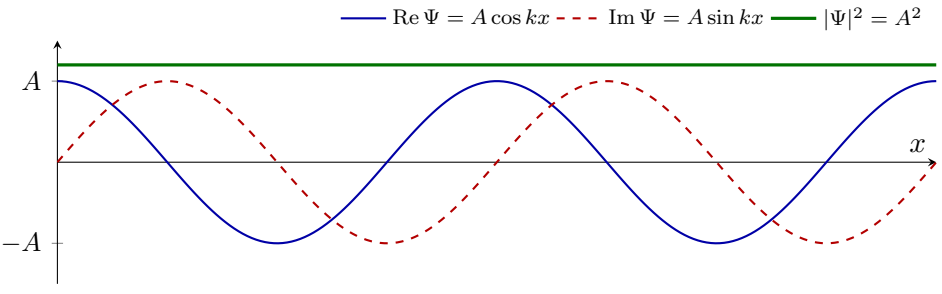


Figure 1: The plane wave $\Psi = Ae^{i(kx-\omega t)}$ at $t = 0$. The real and imaginary parts are a cosine and a sine, a quarter cycle out of phase. The probability density $|\Psi|^2$ (drawn in arbitrary units) is the same everywhere.

Probability density of a plane wave.

$$|\Psi|^2 = \Psi^*\Psi = \left[Ae^{-i(kx-\omega t)}\right]\left[Ae^{+i(kx-\omega t)}\right] = A^2. \tag{20}$$

The probability density is the same at every x and every t. A particle described by a plane wave has a perfectly definite momentum $p = \hbar k$, but it is *equally likely to be found anywhere*. A plane wave is not a realistic description of a localized particle. Its importance is that more realistic wave functions can be built as sums of plane waves of different k. That is the subject of the next lecture.

Why not $A\sin(kx - \omega t)$**?** Taking two x-derivatives of a sine gives back a sine, but one t-derivative gives a cosine, so a real sine or cosine cannot satisfy the Schrödinger equation. The complex exponential works because both derivatives return the same function.